

Scott Ruiter

Advanced Vision / Machine Vision Engineer

Manufacturing Automation | 2D/3D Metrology | Computer Vision | Applied AI

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PROFESSIONAL SUMMARY

Advanced vision and manufacturing automation engineer with 15+ years of experience building production-ready inspection, metrology, and automation systems for electronics manufacturing. Specialized in Cognex VisionPro/Designer, Keyence LJ-X8080/DSCMax 3D laser line-scan, C#/Python workflows, Fuzion machine integration, and high-density connector inspection. Known for converting difficult manufacturing geometry into stable, measurable inspection results across global deployments supporting precision connector and next-generation compute hardware production.

IMPACT SNAPSHOT

- Builds and deploys 2D/3D machine vision systems that combine commercial vision platforms, laser metrology, custom scripts, machine data, and production handoff requirements.
- Advanced fixture-less coordinate metrology using CAD-driven nominal models, measured XY/Z data, and rigid-transform alignment to reduce fixture dependence and setup variance.
- Supports revenue-critical customer programs across global electronics manufacturing environments, including high-density interconnects and AI/compute hardware infrastructure.
- Bridges applications engineering, software development, machine integration, field escalation, customer validation, training, and manufacturing readiness.

TECHNICAL CORE

Machine Vision & Metrology: Cognex VisionPro, Cognex Designer, ToolBlocks, AVP 2D/3D, CNL search, blob analysis, centroid refinement, edge/feature localization, lighting optimization, pin-tip inspection, radial tolerance validation, fixture-less coordinate metrology, CAD-driven nominal models, best-fit alignment

3D Laser Line-Scan: Keyence LJ-X8080, DSCMax, point-cloud processing, Z-height extraction, plane fitting, 3D peak detection, calibrated height profiles, scan-trigger synchronization, linear-scale X/Y tracking, MFOV merging, 3D scan debugging

Manufacturing Automation: SMT/THT automation, Fuzion platform, Fuzion 4.x RG, precision placement, machine troubleshooting, calibration workflows, board-handling offsets, IL7Pick/Place alignment, GEM/host communication, traceability, FDC, PDCA, FAT/SAT

Software, AI & Data: C#, Python, VisionPro scripting, machine-control support, CSV generation, STP feature extraction, OpenCV-oriented workflows, data visualization, debugging utilities, applied machine-learning research, AI-assisted development, RAG knowledge systems

SELECTED ADVANCED VISION PROJECTS

- **AVP 2D/3D Inspection Platform:** Developed production workflows combining Cognex VisionPro, Keyence 3D line-scan, CAD nominal geometry, and coordinate metrology for high-density connector inspection.
- **Keyence LJ-X8080 3D Line-Scan Integration:** Integrated calibration, scan triggering, height extraction, point-cloud processing, plane fitting, Z-peak tagging, and metrology output for pin tips and fine-pitch features.
- **Fixture-less Coordinate Metrology:** Built CAD-driven CSV nominal models and best-fit rigid-transform workflows using measured XY/Z feature data to reduce fixture dependence and setup variance.
- **Paladin Connector Inspection:** Improved pin detection, radial tolerance validation, CNL/blob pairing, lighting strategy, pitch behavior, row alignment, and pin-to-pin interaction handling.

PROFESSIONAL EXPERIENCE

Universal Instruments Corporation - New York, United States

Nov 2018 - Present

Application Engineer III - Advanced Vision, 3D Metrology & Automation

- Develop and deploy production-ready AVP 2D/3D inspection systems for high-density connector, SMT automation, precision placement, and compute-hardware manufacturing programs across Celestica, Cisco, Amazon, NVIDIA, Fabrinet, Ingrasys, Amphenol, Raytheon, and Arista engagements.
- Integrate Keyence LJ-X8080 and DSCMax 3D laser line-scan sensors with Cognex VisionPro workflows, including calibration, scan triggering, point-cloud processing, plane fitting, Z-height extraction, and 3D peak detection.
- Architect fixture-less coordinate-metrology methods using CAD-driven CSV nominal models, measured XY/Z features, and rigid-transform alignment to reduce dependence on fixtures, static models, and fiducials.
- Stabilize and expand Cognex VisionPro inspection pipelines using CNL search, blob extraction, centroid refinement, pattern-free alignment, lighting strategy, and high-accuracy feature localization.
- Automate AVP and 3D scanner workflows for MFOV data saving, pin-set extraction, best-fit alignment, Z-peak tagging, CSV generation, grid visualization, and inspection debugging.

- Deliver VisionPro C# and Fuzion software improvements, including AVPForm1 updates, CSV batch handling, ToolBlock simplification, robust CNL/blob logic, and cross-branch feature support.
- Support Fuzion 4.x RG machine integration involving machine communication, board-handling offsets, calibration, trim workflows, pick/place alignment, traceability, FDC, and PDCA requirements.
- Resolve production and field issues involving bent-pin tolerance behavior, pitch outliers, pattern-search stability, lighting inconsistencies, inspection false failures, and customer acceptance risk.
- Train field engineers, manufacturing teams, training-center personnel, and customer teams on AVP setup, calibration, troubleshooting, and inspection readiness; contribute to patent-level fixture-less 2D/3D metrology concepts.

Selected impact: Expanded advanced inspection capability from traditional 2D vision into combined 2D/3D coordinate metrology; improved scalability by reducing fixture dependence and setup variance; helped replace limited 3D workflows with a practical Keyence/Cognex hybrid architecture; supported factory-ready inspection systems and rapid escalations across multiple global manufacturing sites.

Independent AI, Data & Blockchain Systems - United States

Apr 2013 - Present

Founder / Independent Developer

- Build AI-assisted workflows for software development, RAG knowledge systems, manufacturing intelligence, predictive event analysis, computer-vision concepts, and technical decision support.
- Develop Python-based data-analysis, visualization, automation, and debugging tools aligned with inspection, manufacturing, and applied AI workflows.
- Prototype algorithmic modeling, automated trading strategies, signal-generation logic, backtesting frameworks, and blockchain/Web3 concepts as independent R&D.

Ram Electronics Inc. - Fruitport, MI

Jun 2013 - Sep 2018

SMT / THT Equipment & Process Leader; SMT Equipment Leader / Programmer

- Led SMT/THT equipment programming, process engineering, production optimization, and manufacturing software support for high-mix PCBA production.
- Supported dozens of NPI builds across SMT/THT phases, including placement routing, PPAP support, production ramp-up, quality review, and process validation.
- Optimized process flow, material movement, machine setup, production changeovers, reflow profiling, quality audits, scheduling, troubleshooting, and equipment uptime.
- Maintained software-supported processes aligned with ISO 9001, J-STD-001 Class 2/3 PCBA standards, and DFMEA practices; supported Windows Server, SQL Server, IIS, Active Directory, and related IT systems.

EDUCATION & CERTIFICATIONS

Bachelor's Degree, Computer Science - Baker College Center for Graduate Studies | **Associate's Degree,**

Business/Management/Marketing - Baker College of Muskegon

Certifications & honors: CompTIA A+ | J-STD-001 | Entry Level R | Introduction to R | Intro to Statistics with R | Dean's List | Spanish, Elementary